Supporting Information

Investigating Electrode Calendering and its Impact on Electrochemical Performance by Means of a New Discrete Element Method Model: Towards a Digital Twin of Li-Ion Battery Manufacturing

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Contents

S1. NMC particle size distribution and slurry validation

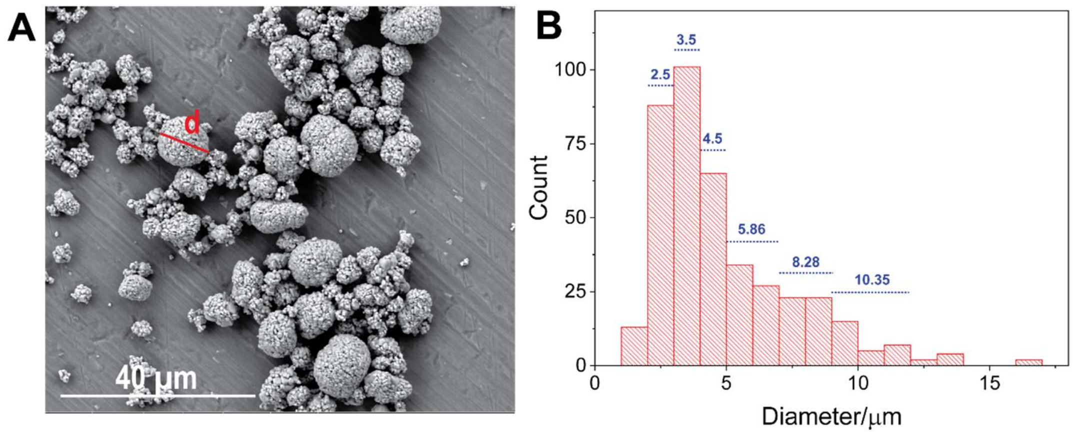

Figure S1. A) Example of Scanning Electron Microscopy (SEM) image used to measure the particle size distribution of NMC. B) Experimental particle size distribution of the NMC. In blue, the six Active Material (AM) particle diameters used for the Coarse Grained Molecular Dynamics (CGMD) models.

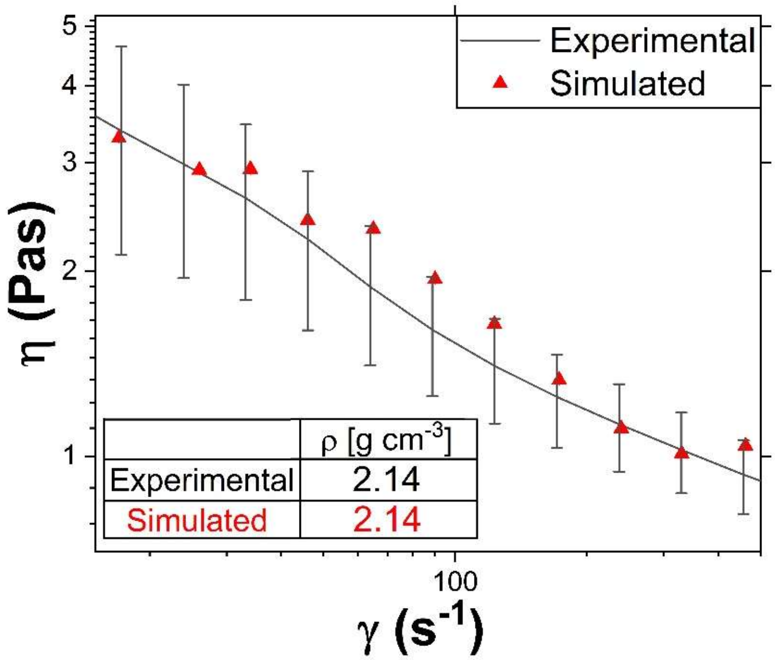

Figure S2. Comparison of experimental and simulated slurry properties for the case of the composition (AM:C65:PVdF) 96:2:2, SC=69%. Lines represent the experimental η-$\dot{\gamma}$ curves, while dots, the simulated data. The $\dot{\gamma}$ range (from tens to hundreds of Hertz) was selected as the one of interest for industrial scale processing. The error bars represent the experimental confidence intervals with a confidence level of 95%. The table on the bottom left of the graph reports the experimental and simulated slurry density (experimental standard deviation equal to 0.01).

S2. CGMD Force Fields (FFs) parameters and evolution from the slurry to the dried electrode

Table S1. CGMD FFs parameters values used for the slurry simulation. The meaning of the parameters was discussed already in Ref. S1. These values were obtained to match the experimental shear viscosity curve and slurry density, as discussed in the main manuscript.

	AM	*CBD*
ε [$pg\ \mu m^2\ \mu s^{-2}$]	$0.01 \times d_{AM\_exp}$	$0.001 \times d_{CBD}$
σ [μm]	$0.88 \times d_{AM\_exp}$	1.1
r_c [μm]	$1.14 \times d_{AM\_exp}$	2.2
d [μm]	$1.14 \times d_{AM\_exp}$	6.2
ρ [$g\ cm^{-3}$]	4.65	0.002
k_n [$pg\ \mu m^{-1}\ \mu s^{-2}$]	8	
γ_n [$\mu m^{-1}\ \mu s^{-1}$]	30	
v	0.15	
X_u	0.015	

Table S2. CGMD FFs parameters values used for the dried electrode simulation. The meaning of the parameters was discussed already in Ref. S1. These values were obtained to match the experimental electrode density and porosity, as discussed in the main manuscript.

	AM	*CBD*
ε [$pg\ \mu m^2\ \mu s^{-2}$]	$3000 \times d_{AM\_exp}$	$3500 \times d_{CBD}$
σ [μm]	$0.88 \times d_{AM\_exp}$	1.1
r_c [μm]	$1.14 \times d_{AM\_exp}$	2.2

d [μm]	$1.02 \times d_{AM\_exp}$	1.3
ρ [g cm^{-3}]	4.65	0.95
k_n [pg μm^{-1} μs^{-2}]	500.0	
γ_n [μm^{-1} μs^{-1}]	10.0	
v	0.3	
X_u	15	

ε, σ and r_c represent the depth of potential well, the distance at which the inter-particle potential is zero and the cut-off of the Lennard-Jones potential. d, ρ, d_{AM_exp} represent the particles diameter, density and experimental diameter of AM particles (blue lines, Figure S1 B). The diameter of AMs particles is slightly expanded respect their experimental counterpart (d_{AM_exp}) to consider the solvent around AM particles (for the case of the slurry) and to allow a certain degree of overlap, needed for the Granular-Hertzian (GH) to be non-zero (as discussed in the reference S1 and in the main manuscript). k_n γ_n, v and X_u represent the elastic constant and the damping factor in the normal direction, the Poisson ratio and the coefficient of friction, respectively. The ratio between the tangential and normal elastic constant and damping factor was calculated as a function of the Poisson ratio (supporting information of the reference S1). The values of the ratios k_t/ k_n and γ_t/γ_n used herein are respectively 1.378 and 0.959 for the slurry and 1.235294118 and 0.907485213 for the dried electrode.

S3. Effect of electrode surface area on the indentation curve

Figure S3 reports the effect of the simulated electrode surface area on the indentation curve. It can be noticed that for small surface area (green curve) the indentation curve is noisy, while the fluctuations decrease by increasing the surface area of the simulated electrode (purple and light blue curves). This observation is understandable considering the dimension of AM particles (~3-10 μm) respect the electrode surface area and taking into account that the forces measured along the indentation are computed as the forces between the particles and the moving plane. Indeed, if the electrode surface is too small, the movement of a single big AM particle can lead to a significant variation in the computed force, giving rise to a noisy indentation curve. On the contrary, a bigger electrode surface mitigates this issue due to the increased number of particles in contact with the moving plane.

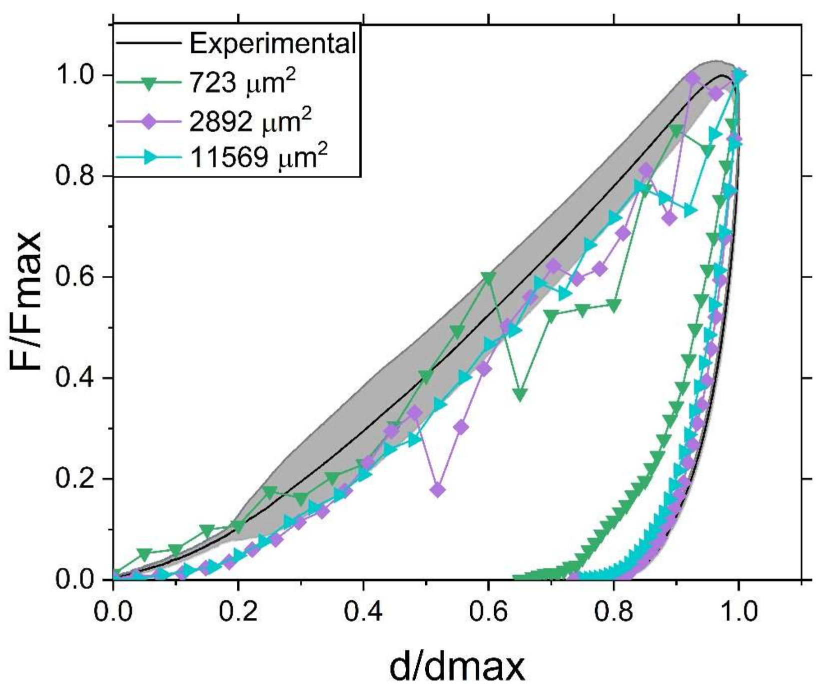

Figure S3. Comparison of experimental and simulated indentation curves (electrode composition: NMC 96 wt%, carbon black 2 wt% and PVdF2 wt%) for different simulated electrodes surface areas. The bigger electrodes were obtained by replicating the simulated electrode coming from the CGMD simulations (green curve) by a factor of 2 (purple curve) or 4 (light blue curve) in the x and y directions.

Overall, a significant effect of size can be observed comparing the green (723 μm^2) and light blue curves (2892 μm^2), while going from the latter to the purple curve (11569 μm^2) the indentation curve is smoother, but the overall behaviour is maintained. Consequently, the use of a dried electrode having a surface of 11569 μm^2 was chosen as the best compromise to avoid errors coming from the size of the simulation box used.

S4. Cross-validation of the pore size distribution algorithm

Considering the approximations used for any algorithm aiming at measuring the pore size distribution (PSD) of an electrode, a second algorithm strategy[2] was tested in order to verify the results shown in Figure 3. This approach considers both spherical and cylindrical pores, allowing to extract the PSD of both topologies, as reported in Figure S4.

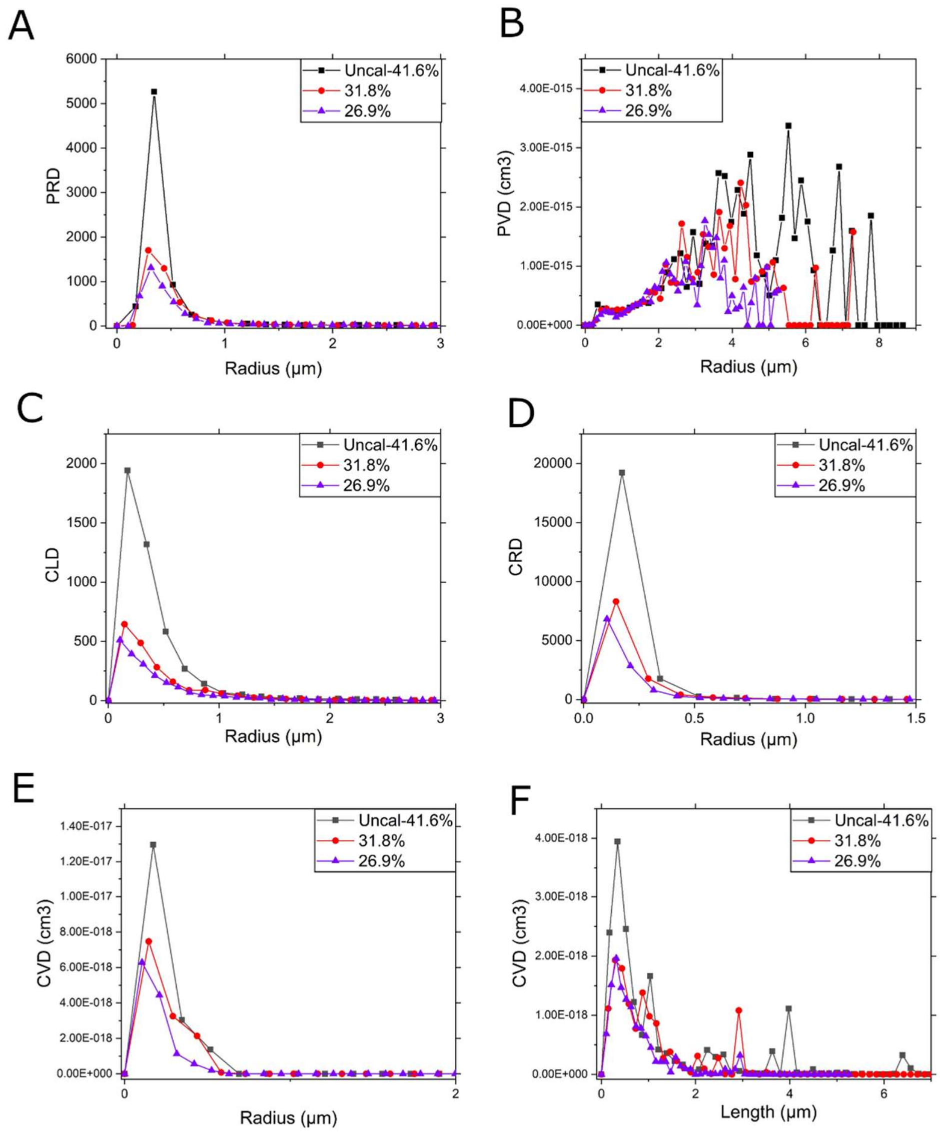

Figure S4. A) Spherical pore radius distribution as a function of sphere radius. B) Spherical pore volume distribution as a function of sphere radius. C) Cylindrical length distribution as a function of cylinder length. D) Cylindrical radius distribution as a function of cylinders radius. E) Cylindrical volume distribution as a function of the cylinder radius. F) Cylindrical volume distribution as a function of the cylinder length.

Overall, the results displayed in Figure S4 confirm the results reported and discussed in the main manuscript in terms of the evolution of the PSD vs. the applied calendering pressure, *i.e.* the pores (both spherical and cylindrical) amount and dimensions decrease by increasing the calender pressure.

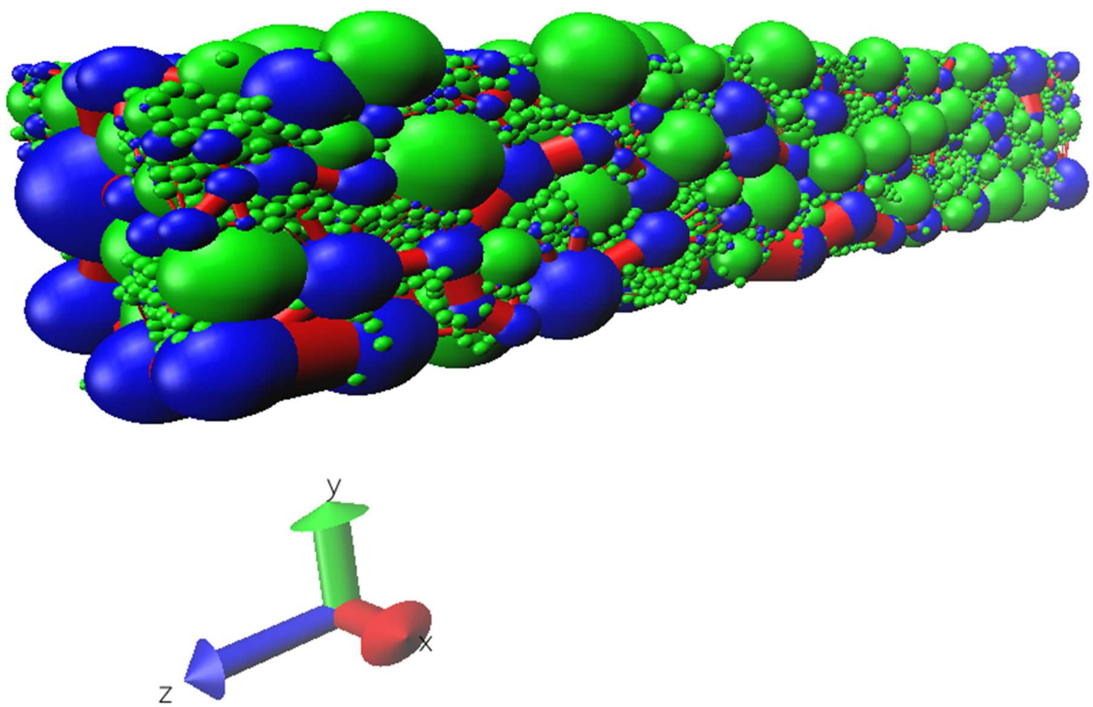

Figure S5. Example of pore network extraction for the case of the calendered electrode having a porosity of 31.5%. Green: solid particles (AM + CBD); blue: spherical pores; red: cylindrical pores.

S5. Details on the AM-AM g(r) peaks

The position of all the principal peaks in Figure 4 C in the main manuscript together with the combination(s) of AM particle radii which can explain the position of each of them is reported in Table S3. Clearly, other combinations involving CBD particles separating AM ones are expected to play a role as well in the peaks position and size.

Table S3. Principal peaks analysis (Figure 4 C in the main manuscript) in terms of their position and the associated AM particle types associated to them.

Peak position [μm]	AM types combination
2.55	AM1 + AM1
3.05	AM1 + AM2
3.57	AM1 + AM3 / AM2 +AM2
4.07	AM2 + AM3
4.77	AM2 + AM4
5.49	AM1 + AM5
6.01	AM2 +AM5
6.51	AM1 + AM6 / AM3 +AM5
7.21	AM4 +AM5
7.59	AM3 + AM6
8.43	AM5 + AM5
9.51	AM5 +AM6

S6. Effect of plane speed on the calendered electrodes

In this section, the effect of the calendering plane speed on the electrode pore size distribution and radial distribution function is reported. Particularly, Figures S6-S7 show (for the case example of the calendered electrode having a porosity of 31.5%) that the plane speed used here (2×10^{-3} µm/µs) is slow enough to capture the evolution of the system through our DEM model, considering that slower speeds do not significantly influence the electrode mesostructure.

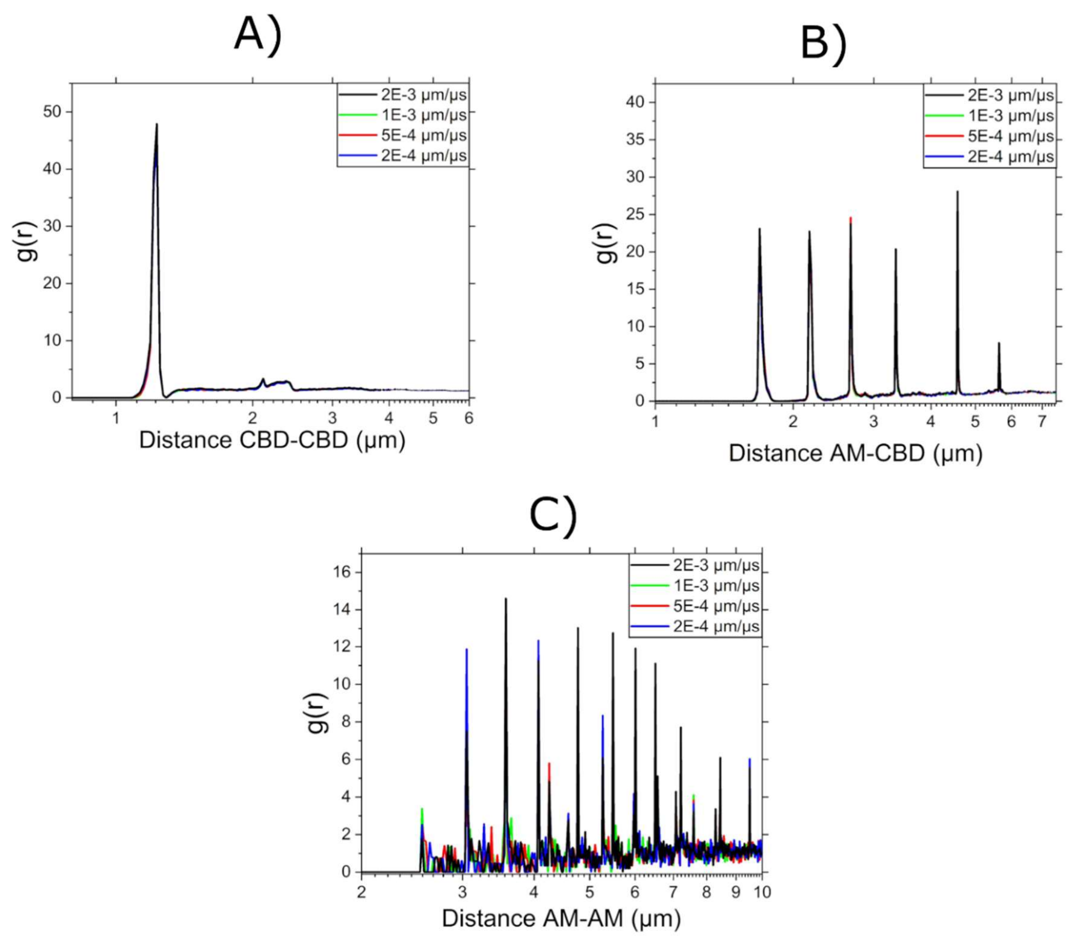

Figure S6. Radial distribution function of the calendered electrode having a porosity of 31.5%, which was calendered at different plane speeds for A) CBD-CBD, B) CBD-AM and C) AM-AM particles. The distances are calculated from the center of mass of A) CBD, B) AM and C) AM particles.

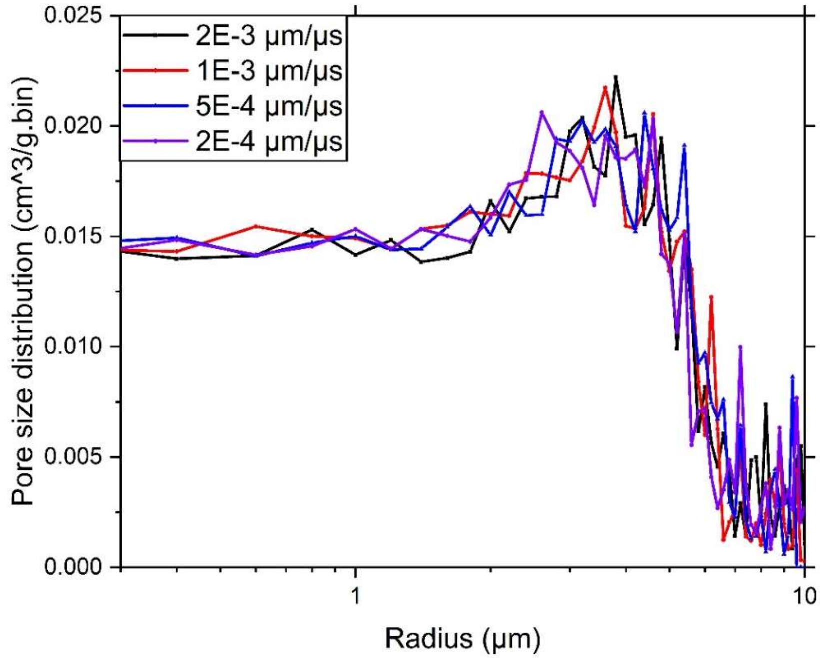

Figure S7. PSD of simulated electrodes (normalized by its mass) having a porosity of 31.5% and calendered at different plane speeds. The PSD were calculated by the open source software PorosityPlus[3], which was adapted to our case study.

S7. Experimental discharge curves

For comparative purposes, in this section we present the experimental discharge curves of NMC-CB-PVdF electrodes (96-2-2), calendered at approximately the same pressures as the simulated ones (Figure 7B). Figure S8 shows the average discharge curves at a C-rate equal to 1C (theoretical specific capacity = 177 mAh g^{-1}, average of 3 different experiments) for electrodes calendered at pressures equal to 5.92 MPa (red line, porosity = 32 ± 1) and 86 MPa (blue line, porosity = 26.3 ± 0.8). The un-calendered electrode shows almost zero capacity at that C-rate.

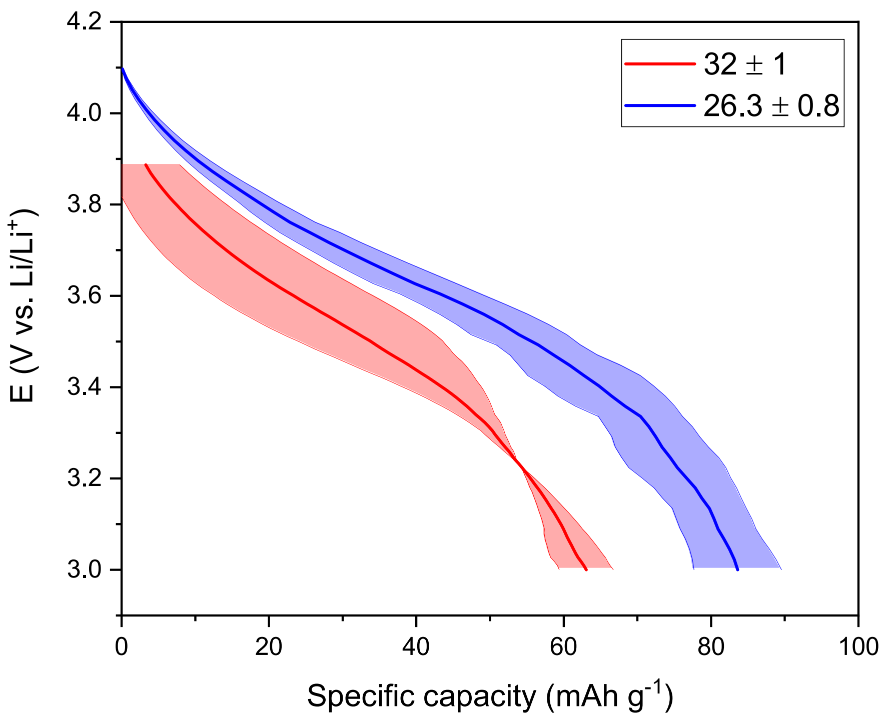

Figure S8. Average discharge curves (performed at 1C rate) for NMC-CB-PVdF electrodes (96-2-2) with porosities equal to 32 ± 1 (red line, applied calender pressure = 5.92 MPa) and 26.3 ± 0.8 (blue line, applied calender pressure = 86 MPa). The shaded areas below and above each curve correspond to the standard deviation of the averages.

S8. References

(S1) Lombardo, T.; Hoock, J.; Primo, E.; Ngandjong, C.; Duquesnoy, M.; Franco, A. A. Accelerated Optimization Methods for Force-Field Parametrization in Battery Electrode Manufacturing Modeling. *Batter. Supercaps* **2020**. https://doi.org/10.1002/batt.202000049.

(S2) Torayev, A.; Rucci, A.; Magusin, P. C. M. M.; Demortière, A.; De Andrade, V.; Grey, C. P.; Merlet, C.; Franco, A. A. Stochasticity of Pores Interconnectivity in Li-O2 Batteries and Its Impact on the Variations in Electrochemical Performance. *J. Phys. Chem. Lett.* **2018**, *9* (4), 791–797. https://doi.org/10.1021/acs.jpclett.7b03315.

(S3) https://research.csiro.au/mmm/porosityplus/ (accessed on March 2020)